CampusLaundry Management System

Database Triggers Specification & Reactive Integrity Enforcement

Project: CampusLaundry Management System	**Category:** Database Triggers & Business Rules	**DBMS Engine:** MySQL 8.0+ / InnoDB
Schema Normalization: 3NF (Strict Third Normal Form)	**Academic Scope:** Database Management Systems Laboratory	**Deployment:** Cloud Production Verified (Aiven)

1. Event-Driven Database Automation Architecture

Database triggers provide declarative, low-level operational guardrails that cannot be bypassed by buggy client code, manual SQL interfaces, or unauthorized API endpoints. In CampusLaundry, triggers automate five vital business responsibilities: (1) Strict REGEXP validation on student onboarding, (2) Update integrity checks, (3) Dynamic turnaround SLA scheduling, (4) Automated order completion timestamps and cancellation quota refunds, and (5) Non-blocking administrative login audits.

2. Trigger 1: trg_validate_student_insert (Onboarding Validation)

Timing: BEFORE INSERT ON students
Enforced Rules: Exactly 10-digit numeric Student ID, 10-digit phone number format, valid hostel block code (B1 to B12 or G1 to G8), and valid semester quota allowance (15 or 30). Initializes default cycle balances automatically.

```
DELIMITER //

DROP TRIGGER IF EXISTS trg_validate_student_insert //

CREATE TRIGGER trg_validate_student_insert
BEFORE INSERT ON students
FOR EACH ROW
BEGIN
    -- 1. Validate Student ID format (10 numeric digits)
    IF LENGTH(NEW.std_id) != 10 OR NEW.std_id NOT REGEXP '^[0-9]{10}$' THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Student ID must be exactly 10 numeric digits.';
    END IF;

    -- 2. Validate Contact Number (10 numeric digits)
    IF LENGTH(CAST(NEW.contact AS CHAR)) != 10
       OR CAST(NEW.contact AS CHAR) NOT REGEXP '^[0-9]{10}$' THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Phone number must be exactly 10 digits.';
    END IF;

    -- 3. Validate Hostel Code (B1-B12 for Boys, G1-G8 for Girls)
    IF NEW.hostel NOT REGEXP '^(B([1-9]|1[0-2])|G[1-8])$' THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Invalid hostel. Allowed hostels are B1 to B12 and G1 to G8 only.';
    END IF;

    -- 4. Validate Semester Quota (Permitted options: 15 or 30 cycles)
    IF NEW.quota NOT IN (15, 30) THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Quota cycles must be either 15 or 30.';
    END IF;

    -- 5. Auto-Initialize Remaining & Used Balances
    IF NEW.cycle_remaining IS NULL THEN
        SET NEW.cycle_remaining = NEW.quota;
    END IF;
    IF NEW.cycle_used IS NULL THEN
        SET NEW.cycle_used = 0;
    END IF;
END //

DELIMITER ;
```

Violation Test Scenario:

```
-- Attempting insertion with invalid phone format (e.g. 5 digits) throws SQLSTATE 45000:
INSERT INTO students (std_id, std_name, contact, hostel, room, academic_year, quota)
VALUES ('2502050999', 'Test Student', 12345, 'H99', '101', '1st Year', 30);
-- Result: ERROR 1644 (45000): Validation Error: Phone number must be exactly 10 digits...
```

3. Trigger 2: trg_validate_student_update (Update Integrity)

Timing: BEFORE UPDATE ON students
Enforced Rules: Ensures that updates made via administrative portals or student profile edits cannot circumvent format constraints on ID, phone numbers, hostel codes, or quota values.

```
DELIMITER //

DROP TRIGGER IF EXISTS trg_validate_student_update //

CREATE TRIGGER trg_validate_student_update
BEFORE UPDATE ON students
FOR EACH ROW
BEGIN
    IF LENGTH(NEW.std_id) != 10 OR NEW.std_id NOT REGEXP '^[0-9]{10}$' THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Student ID must be exactly 10 numeric digits.';
    END IF;

    IF LENGTH(CAST(NEW.contact AS CHAR)) != 10
       OR CAST(NEW.contact AS CHAR) NOT REGEXP '^[0-9]{10}$' THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Phone number must be exactly 10 digits.';
    END IF;

    IF NEW.hostel NOT REGEXP '^(B([1-9]|1[0-2])|G[1-8])$' THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Invalid hostel. Allowed hostels are B1 to B12 and G1 to G8 only.';
    END IF;

    IF NEW.quota NOT IN (15, 30) THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Validation Error: Quota cycles must be either 15 or 30.';
    END IF;
END //

DELIMITER ;
```

4. Trigger 3: trg_set_expected_delivery (Turnaround SLA Calculation)

Timing: BEFORE INSERT ON laundry_transaction
Enforced Rules: Dynamically calculates the promised Expected_Completion_Date. If dropped off on a weekend (Friday or Saturday, when campus laundry facilities encounter peak load surges), a **5-day SLA** is assigned. On regular weekdays (Sunday through Thursday), a **3-day turnaround SLA** is scheduled.

```
DELIMITER //

DROP TRIGGER IF EXISTS trg_set_expected_delivery //

CREATE TRIGGER trg_set_expected_delivery
BEFORE INSERT ON laundry_transaction
FOR EACH ROW
BEGIN
    DECLARE v_day_of_week INT;

    -- Default drop-off date to CURRENT_DATE if omitted by client
    IF NEW.Drop_Off_Date IS NULL THEN
        SET NEW.Drop_Off_Date = CURRENT_DATE();
    END IF;

    SET v_day_of_week = DAYOFWEEK(NEW.Drop_Off_Date);

    -- MySQL DAYOFWEEK: 6 = Friday, 7 = Saturday (Weekend Peak Load Window)
    IF v_day_of_week IN (6, 7) THEN
        SET NEW.Expected_Completion_Date = DATE_ADD(NEW.Drop_Off_Date, INTERVAL 5 DAY);
    ELSE
        SET NEW.Expected_Completion_Date = DATE_ADD(NEW.Drop_Off_Date, INTERVAL 3 DAY);
    END IF;
END //

DELIMITER ;
```

Verification Query:

```
-- Drop-off on a Friday automatically receives a 5-day delivery window:
INSERT INTO laundry_transaction (Transaction_ID, std_id, Drop_Point_ID, Drop_Off_Date, Total_Weight)
VALUES ('TXN9999', '2502050680', 'DP01', '2026-10-30', 5.20);

SELECT Transaction_ID, Drop_Off_Date, DAYNAME(Drop_Off_Date) AS day_name, Expected_Completion_Date
FROM laundry_transaction WHERE Transaction_ID = 'TXN9999';
-- Output: 2026-10-30 | Friday | 2026-11-04 (Exactly 5 days later)
```

5. Trigger 4: trg_before_update_laundry_txn (Completion & Refund Automation)

Timing: BEFORE UPDATE ON laundry_transaction

Enforced Rules:

1. Automatically stamps Actual_Completion_Date = CURRENT_DATE() the moment an order transitions to 'Ready for Pickup' or 'Completed'.
2. If an order status transitions to 'Cancelled', it automatically restores the deducted cycle quota to the student's account (cycle_remaining = cycle_remaining + 1, cycle_used = cycle_used - 1) atomically.

```
DELIMITER //

DROP TRIGGER IF EXISTS trg_before_update_laundry_txn //

CREATE TRIGGER trg_before_update_laundry_txn
BEFORE UPDATE ON laundry_transaction
FOR EACH ROW
BEGIN
    -- 1. Automatic Completion Timestamp
    IF NEW.Status IN ('Ready for Pickup', 'Completed') AND OLD.Actual_Completion_Date IS NULL THEN
        SET NEW.Actual_Completion_Date = CURRENT_DATE();
    END IF;

    -- 2. Automatic Quota Refund on Order Cancellation
    IF NEW.Status = 'Cancelled' AND OLD.Status != 'Cancelled' THEN
        UPDATE students
        SET cycle_remaining = cycle_remaining + OLD.Cycles_Deducted,
            cycle_used = GREATEST(0, cycle_used - OLD.Cycles_Deducted)
        WHERE std_id = OLD.std_id;
    END IF;
END //

DELIMITER ;
```

6. Trigger 5: trg_admin_login_timestamp (Security Audit Ledger)

Timing: BEFORE UPDATE ON admin_users
Enforced Rules: Ensures that when an administrator successfully authenticates, their last_login timestamp is guaranteed to be set to current database server time (NOW()), maintaining an immutable forensic audit log.

```
DELIMITER //

DROP TRIGGER IF EXISTS trg_admin_login_timestamp //

CREATE TRIGGER trg_admin_login_timestamp
BEFORE UPDATE ON admin_users
FOR EACH ROW
BEGIN
    IF NEW.last_login IS NOT NULL AND (OLD.last_login IS NULL OR NEW.last_login != OLD.last_login) THEN
        SET NEW.last_login = NOW();
    END IF;
END //

DELIMITER ;
```

7. Trigger Catalog Inspection & Maintenance Queries

```
-- 1. List All Active Triggers in Schema
SHOW TRIGGERS FROM laundry;

-- 2. Comprehensive Metadata Query from Information Schema
SELECT
    TRIGGER_NAME,
    EVENT_MANIPULATION,
    EVENT_OBJECT_TABLE,
    ACTION_TIMING,
    CREATED
FROM information_schema.TRIGGERS
WHERE TRIGGER_SCHEMA = 'laundry'
ORDER BY EVENT_OBJECT_TABLE, ACTION_TIMING;
```